

Functions of many variables – 2

The gradient as an instrument

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Theoretical minimum.

Theorem 1 (mean value along a segment, and its failure). For a real-valued f differentiable on a neighbourhood of the segment $[a, b]$ there is $\theta \in (0, 1)$ with $f(b) - f(a) = \langle \nabla f(a + \theta(b - a)), b - a \rangle$. For maps into $\mathbb{R}^m$, $m \geq 2$, this is false already for $t \mapsto (\cos t, \sin t)$ on $[0, 2\pi]$; what remains is the mean value inequality $\|f(b) - f(a)\| \leq \sup_{[a, b]} \|Df\| \cdot \|b - a\|$, proved by applying the scalar case to $t \mapsto \langle f(a + t(b - a)), w \rangle$ with a fixed unit vector w .

Theorem 2 (vanishing gradient). If $\nabla f \equiv 0$ on a connected *open* set, then f is constant there: connected open sets in $\mathbb{R}^n$ are polygonally connected. On a connected set that is not open, or not path-connected, the statement is still true for a function smooth in a neighbourhood — but the proof can no longer join two points by a path inside the set.

Theorem 3 (Lipschitz gradient: the descent lemma). If ∇f is L -Lipschitz, then

$$|f(y) - f(x) - \langle \nabla f(x), y - x \rangle| \leq \frac{L}{2} \|y - x\|^2.$$

Consequently $\|\nabla f(x)\|^2 \leq 2L(f(x) - \inf f)$, and the gradient step $x \mapsto x - \frac{1}{L} \nabla f(x)$ decreases f by at least $\frac{1}{2L} \|\nabla f(x)\|^2$.

Fact 1 (convexity through the gradient). A differentiable f is convex if and only if $f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle$ for all x, y , and if and only if ∇f is monotone: $\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq 0$. If $f - \varepsilon \|x\|^2$ is convex (f is strongly convex), the inequality is strengthened by $2\varepsilon \|x - y\|^2$.

Fact 2 (the gradient detects a minimum at infinity). If $\langle \nabla f(x), x \rangle > 0$ outside some ball, then f attains its global minimum inside that ball: along every ray leaving the ball the function increases. This is the standard substitute for compactness on $\mathbb{R}^n$.

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1. Prove the mean value theorem along a segment for a real-valued function, show by the example $t \mapsto (\cos t, \sin t)$ that it has no vector-valued analogue, and deduce the mean value inequality $\|f(b) - f(a)\| \leq \sup \|Df\| \cdot \|b - a\|$. Deduce that a map with bounded derivative is Lipschitz on every convex set, and give an example of a domain where this fails.
 2. (МФТИ 2018) Let a smooth function $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ have zero differential at every point of a connected compact set $X \subset \mathbb{R}^2$. Prove that f is constant on X . (Note that connected and path-connected are different things: X is disconnected if it splits into two disjoint non-empty sets, each of the form $X \cap U$ with U open.)
 3. (Капачёв 4.10) Let $B \subset \mathbb{R}^n$ be the closed unit ball centred at the origin and let B' be a larger ball with the same centre. A function $f: B' \rightarrow \mathbb{R}$ is continuously differentiable and $\|\nabla f(x)\| \geq 1$ for every $x \in B$. Prove that $\max_{x \in B} f(x) - \min_{x \in B} f(x) \geq 2$.
 4. (IMC) Find all $r > 0$ such that whenever $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ is a differentiable function with $\|\nabla f(0, 0)\| = 1$ and $\|\nabla f(u) - \nabla f(v)\| \leq \|u - v\|$ for all $u, v \in \mathbb{R}^2$, the maximum of f on the disc $\{u : \|u\| \leq r\}$ is attained at exactly one point.
 5. (МФТИ 2010) Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be continuously differentiable and suppose that for some $\varepsilon > 0$ the function $f(x) - \varepsilon\|x\|^2$ is convex on $\mathbb{R}^n$. Let x_0 be a point of global minimum of f . Prove that $f(x) - f(x_0) \geq \varepsilon\|x - x_0\|^2$ for all $x \in \mathbb{R}^n$.